

Allocation problems for gross substitutes and submodular valuations

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1 Introduction

In these lectures, we consider algorithms for discrete allocation problems. We let V denote a set of m items that need to be divided between a set A of n agents. The utility of agent i over a subset $S \subseteq V$ of items is $v_i(S)$. We consider the problem of finding an allocation $S_1, S_2, \dots, S_n$ that forms a partition of S and maximizes a certain social welfare function. Three common examples are these:

- Utilitarian welfare: $\max_{S_1, \dots, S_n} \sum_{i \in A} v_i(S_i)$.
- Max-min welfare (Santa Claus): $\max_{S_1, \dots, S_n} \min_{i \in A} v_i(S_i)$.
- Nash welfare: $\max_{S_1, \dots, S_n} \prod_{i \in A} v_i(S_i)$.

The talks will focus on the first and third objectives, and focus on the following valuation functions:

- Additive: $v(S) = \sum_{j \in S} v(\{j\})$
- Submodular: $v(S) + v(T) \geq v(S \cap T) + v(S \cup T)$ for all $S, T \subseteq V$
- $M^\natural$ -concave / gross substitutes defined below.

Additive valuations will be given explicitly by the $v(\{j\})$ values. For the other valuations, the standard model is to assume a *demand oracle*. We use $v(T|S) = v(S \cup T) - v(S)$, and $v(i|S) = v(\{i\}|S)$. We shall also assume $v(\emptyset) = 0$.

Definition 1.1. *The function $v : 2^V \rightarrow \mathbb{R}$ is $M^\natural$ -concave if*

$$v(S) + v(T) \leq \max \left\{ v(S - i) + v(T + i), \max_{j \in T \setminus S} v(S - i + j) + v(T + i - j) \right\}, \quad \forall S, T \subseteq V, i \in S \setminus T. \quad (1)$$

To define gross substitutes, we need to introduce the demand set. For a price vector $p \in \mathbb{R}^V$, let

$$D(v, p) := \arg \max_{S \subseteq V} \{v(S) - p(S)\}$$

Definition 1.2. *The function $v : 2^V \rightarrow \mathbb{R}$ satisfies the gross substitutes property, if for any two price vectors $p \leq p'$ and any $S \in D(v, p)$, there exists $S' \in D(v, p')$ such that $S' \supseteq S \cap \{i : p_i = p'_i\}$.*

As a third characterization, let us define the *greedy demand oracle*.

Algorithm 1 Greedy demand oracle

Input: Valuation function $v : 2^V \rightarrow \mathbb{R}$, $p \in \mathbb{R}^n$

```
1:  $S \leftarrow \emptyset$ 
2: while  $\exists j \in V \setminus S$  such that  $v(j|S) > p_j$  do
3:    $j \leftarrow \arg \max_{j \in V \setminus S} v(j|S) - p_j$ 
4:    $S \leftarrow S \cup j$ 
return  $S$ 
```

Definition 1.3. The function $v : 2^V \rightarrow \mathbb{R}$ is matroidal map, if for any $p \in \mathbb{R}^n$ and any tiebreaking, the greedy demand oracle outputs $S \in D(v, p)$.

We will show later that these three characterizations give the same class:

Theorem 1.4. For the function $v : 2^V \rightarrow \mathbb{R}$, the following are equivalent: v is $M^\sharp$ -concave; v satisfies the gross substitutes property; v is a matroidal map.

Gross substitutes were defined by Kelso and Crawford in 1982, and $M^\sharp$ -concave functions by Murota in 1996. The fundamental equivalence of these two notions was revealed by Fujishige and Yang in 2003.

Example 1.5. Let $v(S)$ be the rank function of a matroid $\mathcal{M}$ on v . Verify that all three properties hold.

Example 1.6. Let $v(S) = \max_{j \in S} s_j$ for a given $s \in \mathbb{R}^n$. Verify that all three properties hold.

Example 1.7. Let $A \subseteq (\mathbb{R}[x])^{d \times n}$ be a matrix with n columns and d rows and polynomial entries. For $X \subseteq [n]$, $|X| = d$, let us define $f(X)$ as the degree of the determinant of the submatrix with columns X , and $f(X) = -\infty$ if this is 0 or $|X| \neq d$. This function is also $M^\sharp$ -concave; it was one of the examples of Dress and Wenzel of valuated matroids.

Lemma 1.8. Gross substitutability implies submodularity.

Proof. It suffices to show that for any $S \subseteq V$ and $i, j \notin S$, $v(S + i) - v(S) \geq v(S \cup \{i, j\}) - v(S + j)$. For this, let us pick $M > \max_{k \in V} v(k)$, and set $p_k = -M$ if $k \in S + j$, $p_k = M$ if $k \notin S$. Further, let $p_i = v(i|S + j)$. Thus, $S \cup \{i, j\} \in D(v, p)$. Let us now define p' by $p'_k = p_k$ for $k \neq j$, and $p'_j = M$. Then, there must be a $T \in D(v, p')$ with $S + i \subseteq T$; in fact, we must have $T = S + i$ as the unique choice. Thus, $v(S + i) - p'(S + i) \geq v(S) - p'(S)$, that is, $v(S + i) - v(S) \geq p'_i$. The claim follows by recalling that $p'_i = v(i|S + j) = v(S \cup \{i, j\}) - v(S + j)$. $\square$

2 Maximizing utilitarian social welfare

For additive valuations v_i , maximizing $\sum_{i \in A} v_i(S_i)$ is immediate. Consider now submodular valuations. The problem is already NP-hard for $n = 2$, for valuations of the form $v_1(S) = v_2(S) = \max\{s(S), b\}$, where $b = s(V)/2$.

We can cast it as a matroid constrained submodular maximization problem. Consider the ground set $V^* = V_1 \cup V_2 \cup \dots \cup V_n$, n disjoint copies of V . Let us take the direct sum of the valuation functions $v_1, \dots, v_n$. That is, for $S \subseteq V^*$, let $v(S) = \sum_i v_i(S \cap V_i)$. Define the partition matroid $\mathcal{M}$ on V^* where we can pick at most one item from each V_i . Then, maximizing $\sum_{i \in A} v_i(S_i)$ amounts to

$$\max_{S \in \mathcal{I}} v(S)$$

Theorem 2.1 (Calinescu, Chekuri, Pál, Vondrák [1]). *There is a polynomial time $(1 - 1/e)$ -approximation algorithm for maximizing a monotone nonnegative submodular function over independent sets of a matroid, and consequently, for the utilitarian welfare maximization problem for submodular valuations.*

Theorem 2.2 (Khot, Lipton, Markakis, Mehta [2]). *There is no polynomial time $(1 - 1/e) + \varepsilon$ -approximation for the utilitarian welfare maximization problem for submodular valuations for any $\varepsilon > 0$ unless $P = NP$.*

The algorithm above is the continuous greedy algorithm. A non-optimal, but much simpler constant factor approximation is the greedy allocation:

Lemma 2.3. *The greedy allocation algorithm returns an $1/2$ -approximation for submodular functions.*

Algorithm 2 Greedy allocation

Input: Valuation functions $v_1, v_2, \dots, v_n : 2^V \rightarrow \mathbb{R}$

```
1: for  $i = 1, \dots, n$  do  $S_i \leftarrow \emptyset$   
2: for  $j \in V$  do  
3:    $i \leftarrow \arg \max_{i \in A} v_i(j | S_i)$   
   return  $(S_1, S_2, \dots, S_m)$ 
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Proof. Let $(S_1^*, \dots, S_n^*)$ be an optimal solution. We show that

$$2 \sum_{i=1}^n v_i(S_i) \geq \sum_{i=1}^n v_i(S_i^* \cup S_i). \quad (2)$$

This implies the claim by monotonicity. We show (2) by induction. Numbering $V = \{1, 2, \dots, m\}$ in the order of selection, we show for $j = 0, 1, 2, \dots, n$.

$$2 \sum_{i=1}^n v_i(S_i \cap [j]) \geq \sum_{i=1}^n v_i((S_i^* \cup S_i) \cap [j]). \quad (3)$$

This clearly holds for $j = 0$. Assume we have it for $j - 1$. Let $j \in S_k \cap S_\ell^*$. The greedy choice and monotonicity implies

$$\begin{aligned} v_k(S_k \cap [j]) - v_k(S_k \cap [j - 1]) &\geq v_\ell((S_\ell \cap [j - 1]) + j) - v_\ell(S_\ell \cap [j - 1]) \\ &\geq v_\ell((S_\ell \cup S_\ell^*) \cap [j]) - v_\ell((S_\ell \cup S_\ell^*) \cap [j - 1]). \end{aligned}$$

When moving from $j - 1$ to j , the LHS in (3) increases by $v_k(S_k \cap [j]) - v_k(S_k \cap [j - 1])$, while the RHS by at most

$$v_k((S_k \cup S_k^*) \cap [j]) - v_k((S_k \cup S_k^*) \cap [j - 1]) + v_\ell((S_\ell \cup S_\ell^*) \cap [j]) - v_\ell((S_\ell \cup S_\ell^*) \cap [j - 1])$$

both terms here are bounded by the LHS increase. If $\ell = k$, then the above terms are the same and only need to be counted once. $\square$

2.1 Walrasian market equilibrium

For gross substitute valuations, there is a remarkable connection to Walrasian market equilibrium.

Definition 2.4. Given monotone valuation functions v_i , $i \in A$ on V , we say that the (not necessary complete) allocation $(S_1, S_2, \dots, S_n)$ and prices $p \in \mathbb{R}_{\geq 0}^m$ form a Walrasian equilibrium, if $S_i \in D(v_i, p)$ for all $i \in A$, and $p_i = 0$ for all $i \in V \setminus \cup_{i=1}^n S_i$. For $\varepsilon > 0$, we say that they form a ε -approximate Walrasian equilibrium, if

$$v_i(S_i) - p(S_i) \geq \max_{S \subseteq V} v_i(S) - p(S) - \varepsilon \quad \forall i \in A.$$

The relevance of such an allocation is given by the following observation:

Proposition 2.5. A Walrasian equilibrium allocation $T_1, T_2, \dots, T_n$ maximizes the sum $\sum_{i=1}^n v_i(S_i)$. A ε -Walrasian equilibrium gives a partition with $\sum_{i=1}^n v_i(T_i) \geq \text{OPT} - n\varepsilon$.

Proof. Let p denote the equilibrium price vector, and consider any allocation $S_1, \dots, S_n$. For each agent $i \in A$, $v_i(T_i) - p(T_i) \geq v_i(S_i) - p(S_i)$. Summing up both sides, and using that at equilibrium $p(\sum_{i=1}^n S_i) = p(V)$, we see that

$$\sum_{i=1}^n v_i(T_i) - p(V) \geq \sum_{i=1}^n v_i(S_i) - p(\cup_{i=1}^n T_i),$$

showing the claim. The proof of the approximate variant follows the same way. $\square$

Algorithm 3 Auction algorithm

Input: Valuation functions $v_1, v_2, \dots, v_n : 2^V \rightarrow \mathbb{R}$, $\varepsilon > 0$

```
1:  $\delta \leftarrow \frac{\varepsilon}{m}$ 
2: for  $j \in V$  do  $p_j \leftarrow 0$ 
3: for  $i \in A$  do  $S_i \leftarrow \emptyset$ 
4: Implicitly define for all  $j \in A$ ,  $p^i \in \mathbb{R}_{\geq 0}^V$  by  $p_j^i = p_j$  for  $j \in S_i$  and  $p_j^i = p_j + \delta$  otherwise
5: while  $\exists i \in A : S_i \notin D(v_i, p^i)$  do
6:   Find  $X \supseteq S_i \in D(v_i, p^i)$ 
7:   for  $j \in X \setminus S_i$  do  $p_j \leftarrow p_j + \delta$ 
8:    $S_i \leftarrow X$ ;
9:   for  $k \in A - i$  do  $S_k \leftarrow S_k \setminus X$ 
return  $(S_1, S_2, \dots, S_m)$  and  $p$ 
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Such an allocation may not always exist. However, for gross substitutes valuations, a simple auction algorithm gives an approximate Walrasian equilibrium, and also enables showing existence. This dates back to the 1982 paper by Kelso and Crawford that introduced the gross substitutes class. The demand oracle $D(v_i, p)$ in the algorithm can be implemented by the greedy algorithm.

Lemma 2.6. *In Algorithm 3, it is always possible to select $X \supseteq S_i$. For monotone gross substitute valuations, the algorithm outputs a ε -approximate Walrasian equilibrium, and terminates in $O(m^2 M / \varepsilon)$ iterations, where $M = \max_{i \in A} v_i(V)$.*

Proof. We show by induction that there exists throughout $T_i \in D(v_i, p^i)$ such that $S_i \subseteq T_i$; we can simply pick $X = T_i$. Initially, $T_i = V$ for all $i \in V$ is a suitable choice. The prices p^i can be updated either for an iteration that updates S_i , or an iteration that updates S_k for $k \neq i$. Let us start with the first case. By induction, we can pick $S_i = X = T_i$. This is by definition in $D(v_i, p^i)$ before updating the prices. After the price update, note that $(p')^i = p^i$ still holds, even though $p' \neq p$. Hence, $S_i = T_i$ is a good choice.

Assume now we updated the prices by setting $p_j' = p_j + \delta$ for $j \in X \setminus S_k$, $k \neq i$. For the new set $S_i' = S_i \setminus X$, we have $(p')_j^i = p_j^i$ for $j \in S_i'$, and $(p')^i \geq p^i$. Hence, by the gross substitutes property, there exists some $T_i' \in D(v_i, (p')^i)$ such that $S_i' \subseteq T_i'$.

At termination $S_i \in D_i(v_i, p^i)$ for all $i \in A$, and $p_i = 0$ for $i \in V \setminus (\cup_{i=1}^n S_i)$. It is an ε -approximate equilibrium because for any $S \subseteq V$,

$$v_i(S_i) - p(S_i) = v_i(S_i) - p^i(S_i) \geq v_i(S) - p^i(S) \geq v_i(S) - p(S) - |S|\delta \geq v_i(S) - p(S) - \varepsilon$$

Finally, the bound on the number of iterations follows because the prices must remain bounded by M , and at least one price increases in each iteration. $\square$

Using compactness arguments, it is easy to derive the existence of an exact Walrasian equilibrium.

It is instructive to look at the special cases mentioned above. First, assume each $v_i = r_i$ is the rank function of a matroid. Then, $\max_{S_1, \dots, S_n} \sum_{i=1}^n r_i(S_i)$ amounts to the matroid union problem. Setting $\varepsilon < 1/n$, the algorithm outputs an optimal allocation. Noting also that we can bound $M \leq m$, this runs in a strongly polynomial number of iterations $O(m^3 n)$ each iteration a greedy algorithm.

Next, let $v_i(S) = \max_{j \in S} w_{ij}$ for given values w_{ij} . Then, maximizing $\sum_{i=1}^n v_i(S_i)$ is equivalent to finding a maximum weight matching. The algorithm proceeds as follows. At every bidding step, we select an agent with $S_i = \emptyset$, and let them pick the item maximizing $s_{ij} - p_j - \delta$; they wish to bid when $\max_{j \in V} s_{ij} - p_j - \delta > 0$. Agents currently holding a single item will never bid. At termination, setting $y_i = \max_{j \in V} s_{ij} - p_j$, we have that $y_i + p_j \geq s_{ij}$, along with a matching of edges with $s_{ij} \geq y_i + p_j - \delta$, and all unassigned items having $p_j = 0$, and agents with no assigned goods will have $y_i \leq \delta$. This is the classical auction algorithm for matching by Bertsekas; it is also closely related to preflow-push.

2.2 Properties of gross substitutes functions

Let us first show some of the implications in Theorem 1.4.

Proposition 2.7. *Every $M^\sharp$ -concave function satisfies the gross substitutes property.*

Proof. Consider prices $p \leq p'$, $S \in D(v, p)$, $U = S \cap \{j : p_j = p'_j\}$. Take a set $T \in D(v, p)$ such that $T \cap U$ is maximal. If $U \subseteq T$, we are done. Otherwise, let $i \in U \setminus T$, and apply (1) for S and T . Assume the first bound is valid, that is,

$$v(S) + v(T) \leq v(S - i) + v(T + i).$$

Using that $p_i = p'_i$, and that $S \in D(v, p)$, $T \in D(v, p')$, we get

$$v(S) - p(S) + v(T) - p'(T) \leq v(S - i) - p(S - i) + v(T + i) - p'(T + i) \leq v(S) - p(S) + v(T) - p'(T),$$

implying equality throughout. Thus, $T + i \in D(v, p')$, contradicting the choice of T . The argument is similar in the second case, when for $j \in T \setminus S$,

$$v(S) + v(T) \leq v(S - i + j) + v(T + i - j).$$

Observe that $p(S - i + j) + p'(T + i - j) = p(S) + p'(T) + p_j - p'_j \leq p(S) + p'(T)$, and therefore

$$\begin{aligned} v(S) - p(S) + v(T) - p'(T) &\leq v(S - i + j) - p(S - i + j) + v(T + i - j) - p'(T + i - j) \\ &\leq v(S) - p(S) + v(T) - p'(T), \end{aligned}$$

giving contradiction as in the previous case. $\square$

Proposition 2.8. *Every $M^\sharp$ -concave function is a matroidal map.*

Proof. Given prices p , let S be the output of the greedy demand oracle (Algorithm 1). For a contradiction, assume $S \notin D(v, p)$, and let $T \in D(v, p)$ with $|S \Delta T|$ minimal. First, assume $S \subseteq T$. By the stopping rule of the algorithm, $v(i|S) - p_i \leq 0$ for every $i \in T \setminus S$. Gross substitutes implies submodularity, and therefore $v(i|T - i) - p_i \leq 0$, meaning that $v(T - i) - p(T - i) \geq v(T) - p(T)$ and thus $T - i \in D(v, p)$, contradicting the choice of T .

Hence, $S \setminus T \neq \emptyset$; let us select $i \in S \setminus T$ as the first item chosen in the algorithm outside T . Let $U \subseteq S \cap T$ denote the set of elements chosen earlier. We now apply (1) for $U + i$, T and i . First, assume that

$$v(U + i) + v(T) \leq v(U) + v(T + i).$$

This implies

$$\begin{aligned} v(U + i) - p(U + i) + v(T) - p(T) &\leq v(U) - p(U) + v(T + i) - p(T + i) \\ &< v(U) - p(U), \end{aligned}$$

where the inequality follows by the choice of T minimizing $|S \Delta T|$. Consequently, $v(i|U) - p_i < 0$, and hence the algorithm should not have selected item i after U . Next, assume there exists $j \in T \setminus (U + i)$ such that

$$v(U + i) + v(T) \leq v(U + j) + v(T + i - j).$$

Similarly as above,

$$\begin{aligned} v(U + i) - p(U + i) + v(T) - p(T) &\leq v(U + j) - p(U + j) + v(T + i - j) - p(T + i - j) \\ &< v(U + j) - p(U + j), \end{aligned}$$

and hence j would have been a better choice after U , a contradiction again. $\square$

2.3 Operations on submodular functions

We already checked that matroid rank functions are $M^\natural$ -concave. We now consider operations that preserve this property.

- *Shifting by constant*
- *Multiplication by constant*
- *Deletion:* If $g : 2^V \rightarrow \mathbb{R}_{\geq 0}$ is $M^\natural$ -concave, then for $S \subseteq V$, the function $(g - S) = g|_{(V \setminus S)} : 2^{V \setminus S} \rightarrow \mathbb{R}_{\geq 0}$ defined by $(g - S)(X) = g(X)$ for $X \subseteq V \setminus S$ is also $M^\natural$ -concave.
- *Contraction:* If $g : 2^V \rightarrow \mathbb{R}_{\geq 0}$ is $M^\natural$ -concave, then for $S \subseteq V$, the function $(g/S) : 2^{V \setminus S} \rightarrow \mathbb{R}_{\geq 0}$ defined by $(g/S)(X) = g(X \cup S) - g(S)$ for $X \subseteq V \setminus S$ is also $M^\natural$ -concave.
- *Dual:* If $g : 2^V \rightarrow \mathbb{R}_{\geq 0}$ is $M^\natural$ -concave, then $g^* : 2^V \rightarrow \mathbb{R}_{\geq 0}$ defined as $g^*(X) = g(V \setminus X) + |V \setminus X|$ is also $M^\natural$ -concave.
- *Direct sum*
- *Principal extension:* see below.

Lemma 2.9 (Principal extension). *If $g : 2^V \rightarrow \mathbb{R}_{\geq 0}$ is $M^\natural$ -concave, and $w \in (\mathbb{R} \cup \{-\infty\})^V$, then the principal extension $g^w : 2^{V+r} \rightarrow \mathbb{R}_{\geq 0}$ defined as $g^w(X) = g(X)$ if $X \subseteq V$*

$$f^w(X) = \max_{v \in V \setminus X} \{f(X - r + v) + w_v\}, \quad r \in X \subseteq V$$

is $M^\natural$ -concave.

Proof. Case checking from the axioms. Immediate if $r \notin X, Y$. Assume $r \in X \setminus Y$; let $f^w(X) = f(X - r + v) + w_v$. If $v \notin Y$, then the $M^\natural$ -concave axiom for $X - r + v, Y, v$ gives either

$$f(X - r + v) + f(Y) \leq f(X - r) + f(Y + v),$$

in which case

$$f^w(X) + f^w(Y) = f(X - r + v) + w_v + f(Y) \leq f^w(X - r) + f^w(Y + r),$$

or $\exists u \in Y \setminus (X - r + v)$ such that

$$f(X - r + v) + f(Y) \leq f(X - r + u) + f(Y + v - u),$$

in which case

$$f^w(X) + f^w(Y) = f(X - r + v) + w_v + f(Y) \leq f^w(X - r + u) + f^w(Y + r - u).$$

The other cases can be verified similarly. □

Iteratively applying this simple operation allows for a powerful construction of *induction by bipartite graph*. Let $G = (V, U; E)$ with edge weights $w \in \mathbb{R}^E$, and $g : 2^U \rightarrow \mathbb{R}_{\geq 0}$ an $M^\natural$ -concave function. Let us define $f : 2^U \rightarrow \mathbb{R}_{\geq 0}$ so that

$$f(X) = \max \left\{ g(\sigma(X)) + \sum_{i \in X} w_{i\sigma(i)} : \sigma : X \rightarrow U \text{ injective} \right\}.$$

Lemma 2.10. *The function f defined by induction by network is $M^\natural$ -concave.*

Proof. We create a function $\bar{f}$ on the ground set $\bar{V} = U \uplus V$ by adding the elements of V one-by-one using principal extension of g . One can see by induction that for $X \subseteq V$, $\bar{f}(X)$ is precisely what we need, and then we delete U . □

This can be seen as a generalization of Rado's theorem:

Theorem 2.11 (Rado 1942). *Let $G = (V, U; E)$ be a bipartite graph and $\mathcal{M}$ a matroid on U . Consider the family of subsets $X \subseteq V$ such that X can be matched to an independent subset of U . These form the independent sets of a matroid on V .*

Example 2.12. We get weighted matroid rank functions $r_w(X)$ being the max weight independent set in X by using this for the $0, -\infty$ indicator function of the independent sets.

Two further important corollaries:

Lemma 2.13. *If f and g are $M^\natural$ -concave on the same ground set V , then the max convolution*

$$(f \star g)(X) = \max_{Y \subseteq X} f(Y) + g(X \setminus Y)$$

is also $M^\natural$ -concave.

Proof. We create $U = V' \uplus V''$ and take the direct sum of f and g . Then, we add a bipartite graph connecting each $v \in V$ to its two copies by the edges (v, v') and (v, v'') with 0 cost edges. $\square$

Example 2.14. Rank function of matroid union.

A seemingly more general construction is *induction by network*. Given a directed graph $D = (T, E)$ with edge weights $w \in \mathbb{R}^E$ and two node sets $U, V \subseteq T$ and an $M^\natural$ -concave function g on V , let

$$f(X) = \max \left\{ g(S) + \sum_{e \in \mathcal{P}} w_e : S \subseteq U, \text{ node-disjoint paths } \mathcal{P} \text{ between } X \text{ and } S \right\}.$$

Theorem 2.15. *Every function that can be obtained by induction by network can be obtained by induction by bipartite graph and contraction.*

This is similar to the argument showing that gammoids are minors of transversal matroids.

2.4 Complete classes of valuations

The above operations naturally raise a question on characterising classes of $M^\natural$ -concave functions. A natural starting family is the $0, -\infty$ indicator functions of matroids. From these, we can use the operations of deletion, contraction, duality, direct sum, and principal extension. In particular, we obtain the following functions:

Definition 2.16. *We call a valuation R -induced if it can be obtained from matroid indicator functions by induction by bipartite graph, and R -minor if it can be obtained by contraction from an R -induced valuation.*

Theorem 2.17. *The set of R -minor valuations is closed under deletion, contraction, duality, direct sum, and principal extension. Moreover, it is the smallest closed class containing matroid indicator valuation functions.*

It is not obvious to find an $M^\natural$ concave function *not* in this class. In fact, the Matroid Based Valuation Conjecture by Ostrovsky and Paes Leme asked whether all monotone gross substitute valuations arise from weighted matroid rank functions by taking minors and max convolution. This would follow if R -minor = $M^\natural$ -concave were true.

Theorem 2.18 (Loho, Smith, Husić, Végh 2024). *Not every $M^\natural$ -concave function is R -minor.*

Definition 2.19. *Let $n \geq 3$, $V = [2n]$, and let $\mathcal{F}_n$ be the following family of functions $h : 2^V \rightarrow \mathbb{R} \cup \{-\infty\}$. We consider the pairs $P_i = \{2i - 1, 2i\}$, and let*

$$\mathcal{H} = \{P_i \cup P_j \mid ij \equiv 0 \pmod{2}\}$$

Let $X^ = \{1, 2, 3, 4\}$, and let*

- $h(X) = 0$, if $X \in \binom{V}{4} \setminus \mathcal{H}$
- $h(X) < 0$, if $X \in \mathcal{H}$
- $h(X^*) < 0$ the unique largest nonzero value
- $h(X) = -\infty$ if $|X| \neq 4$.

Theorem 2.20. For $n \geq 3$, the functions in $\mathcal{F}_n$ are $M^\natural$ -concave but not R -minor.

The construction is inspired by sparse paving matroids, and resemble the Vámos matroid.

2.5 The cone of gross substitutes

In general, for $M^\natural$ -concave functions f, g , their sum $f + g$ is *not* $M^\natural$ -concave. For example, coverage functions arise as sums of rank-1 matroid rank functions. We can define the *cone of gross substitutes*, i.e., the functions that arise as sums of $M^\natural$ -concave functions.

Conjecture 2.21. There exists a universal constant $c > 0$, such that for every monotone nonnegative submodular function f with $f(\emptyset) = 0$, there is set of nonnegative monotone gross substitute functions $g_1, g_2, \dots, g_k$,

$$\sum_{i=1}^k g_i(X) \leq f(X) \leq c \sum_{i=1}^k g_i(X), \quad \forall X \subseteq V.$$

For approximating by a single gross substitute function, Dobzinski, Feige, Feldman, Paes Leme in 2022 showed that one cannot approximate submodular functions pointwise better than $O(\log n / \log \log n)$ by a single gross substitute function.

For the cone of gross substitutes, it is nontrivial even to show that it does not include all submodular functions. This follows by a nice argument using discrete Hessians by Dughmi, Roughgarden, and Yan from 2016.

Question 2.22. Can one give an inequality description of the cone of gross substitutes?

2.6 The indirect utility function

For $f : 2^V \rightarrow \mathbb{R}_{\geq 0}$, we can define the *indirect utility function* $\tilde{f} : \mathbb{R}^V \rightarrow \mathbb{R}$ by

$$\tilde{f}(p) = \max f(X) - p(X).$$

Note that $\tilde{f}(p)$ corresponds to the utility maximization problem $D(f, p)$; we also define it for possibly negative prices. Murota calls the negative of this the Legendre–Fenchel transform.

Example 2.23. If r is the rank function of a matroid, and $p \geq 0$, then $\tilde{f}(p) = \hat{r}(1 - p)$ for the Lovász extension $\hat{r}$.

A centrepiece of discrete convex analysis is the duality between $M^\natural$ -concave and $L^\natural$ -convex functions. The indirect utility function of an $M^\natural$ -concave functions is (polyhedral) $L^\natural$ -convex. To be precise, $L^\natural$ -concavity is defined for integer prices. We give an equivalent definition

Definition 2.24. The function $g : \mathbb{Z}^V \rightarrow \mathbb{R}$ is $L^\natural$ -convex, if for any $p, q \in \mathbb{Z}^V$, $p \not\leq q$, if $Z = \arg \max p_i - q_i$, then

$$g(p) + g(q) \geq g(p - \chi_Z) + g(q + \chi_Z).$$

We say that a function $\mathbb{Z}^V \rightarrow \mathbb{R}$ is *lattice submodular*, if for the lattice operations $(p \wedge q)_i = \min(p_i, q_i)$, $(p \vee q)_i = \max(p_i, q_i)$, we have

$$g(p) + g(q) \geq g(p \wedge q) + g(p \vee q),$$

Lemma 2.25. An $L^\natural$ -convex function is lattice submodular.

Proof. One can easily see that it suffices to show that for any $s \in \mathbb{Z}$ and $i, j \in V$,

$$g(s + \chi_i) + g(s + \chi_j) \geq g(s) + g(s + \chi_i + \chi_j).$$

This follows by the repeatedly applying the definition for $p = s + \chi_i$ and $q = s + \chi_j$, and then the other way around. $\square$

3 Polynomial-time algorithms for Walrasian equilibrium

3.1 The Lyapunov function

Let us now get back to the question of efficient Walrasian equilibrium computation. Let $V_i(p) = \max_{S \subseteq V} v_i(S) - p(S)$ be the indirect utility function of agent i , and define the function

$$L(p) = \sum_{i=1}^n V_i(p) + p(V).$$

Lemma 3.1. *There exists a Walrasian equilibrium with prices p if and only if p is a minimizer of $L(p)$. The minimum of $L(p)$ equals $\max \sum_{i=1}^n v_i(S_i)$.*

Proof. We first show weak duality. Let $(S_1, \dots, S_n)$ be any allocation and p be a set prices. We have

$$v_i(S_i) - p(S_i) \leq V_i(p) \quad \forall i \in [n],$$

hence

$$\sum_{i=1}^n v_i(S_i) \leq L(p)$$

At equilibrium, we have equality throughout, implying the claim. (The existence of equilibria follows by the auction algorithm and limit arguments.) $\square$

Let us now consider the following LP:

$$\begin{aligned} \min \quad & \sum_i v_i + \sum_j p_j \\ & v_i \geq u_i(S) - p(S), \quad \forall i \in [n], \quad \forall S \subseteq V, p \geq 0. \end{aligned} \tag{4}$$

This corresponds to minimizing $L(p)$. The separation oracle corresponds to the greedy demand oracle. Hence, we can solve this by the ellipsoid algorithm. The dual is the configuration LP; the gross substitutes property thus guarantees that the dual has integer solution.

Given optimal prices p_j , a ‘brute force’ way to find an optimal allocation is to build a partial assignment $T_1, T_2, \dots, T_n$. These start empty, and at every stage we use the contracted valuations $\bar{v}_i(X) = v_i(X \cup T_i) - v_i(T_i)$. We try to extend it by adding the next unassigned element k to each of the bundles; we succeed if the optimum decreases by p_k . A nicer way will be discussed next.

3.2 An auction algorithm

Assume all u_i ’s are integer valued functions. It can be shown that integer optimal prices exists. We give a pseudopolynomial auction algorithm, using the more general discrete convex optimization perspective.

The Lyapunov function is the sum of $L^{\natural}$ -convex functions, and is thus also $L^{\natural}$ -convex. From $L^{\natural}$ -convexity, it follows that there exists a unique pointwise minimal minimizer p^* of $g(p)$

Algorithm 4 Steepest descent algorithm

Input: $L^{\natural}$ -convex function $g : \mathbb{Z}^V \rightarrow \mathbb{R}$, starting prices $p^0 \in \mathbb{Z}_{\geq 0}^V$ with the guarantee that $p^0 \leq p^*$

- 1: $p \leftarrow p^0$; $p' \leftarrow M\mathbf{1}$;
 - 2: **while** $g(p) < g(p')$ **do**
 - 3: $p' \leftarrow p$;
 - 4: Compute $S \in \arg \min g(p + \chi_S)$;
 - 5: **if** $g(p + \chi_S) < g(p)$ **then** $p \leftarrow p + \chi_S$
 - return** p
-

Lemma 3.2. *The functions $h(S) = g(p + \chi_S)$ is a submodular set function, and therefore S and T in the algorithm can be obtained by submodular function minimization.*

Applied for the Lyapunov function $L(p)$, one can show that S corresponds to a set of overdemanded items, i.e., these items would be required in more than 1 copies in any allocation of optimal bundles. Thus, the algorithm increases their prices. Computing this sets can also be done by a matroid union algorithm. In a paper with Eickhoff, Neuwohner, Peis, Rieken, Vargas Koch, and V. 2024, we give an efficient push-relabel algorithm for the general multi-item setting.

We now show a bound on the number of steps.

Lemma 3.3. *If $p \leq p^*$ before the update, then $p' = p + \chi_S \leq p^*$ after the update, and moreover, $\|p^* - p'\|_\infty < \|p^* - p\|_\infty$. Consequently, the algorithm terminates in $\|p^* - p^0\|_\infty$ iterations.*

Proof. Let us first show $p' \leq p^*$ after the update. Let $Z = \arg \max p_i^* - p_i$. By $p \leq p^*$, we must have $Z \neq \emptyset$. If $\max p_i^* - p_i = 1$, then by the minimal choice of $g(p + \chi_S)$, we must have $p + \chi_S = p^*$ and the algorithm terminates. Assume for the rest $\max p_i^* - p_i \geq 2$. We claim that $Z \subseteq S$; this implies $\|p' - p^*\|_\infty < \|p - p^*\|_\infty$.

For a contradiction, assume $Z \setminus S \neq \emptyset$. By $L^\natural$ -convexity,

$$g(p + \chi_S) + g(p^*) \geq g(p + \chi_{S \cup Z}) + g(p^* - \chi_{Z \setminus S}) \geq g(p + \chi_S) + g(p^*),$$

and it follows that $\bar{p} = p^* - \chi_{Z \setminus S}$ is also a minimizer. This contradicts the minimal choice of p^* . $\square$

3.3 Proximity and scaling

Theorem 3.4. *Let $g : \mathbb{Z}^V \rightarrow \mathbb{R}$ be $L^\natural$ -convex, $n = |V|$, and assume that p satisfies*

$$g(p \pm 2\chi_S) \leq g(p) \quad \forall S \subseteq V.$$

Then, there exists a minimizer $\bar{p}$ of g over $\mathbb{Z}^V$ with

$$\|p - \bar{p}\|_\infty \leq n.$$

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